Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/10/2013
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	inseob.hahn@jpl.nasa.gov	**Fax:**	FY 818-393-4878
PI Organization Type:	NASA CENTER	**Phone:**	818-354-7999
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	4800 Oak Grove Dr.		
PI Address 2:	79-24		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended for reflight to 9/30/2017 (per PI and U. Israelsson/JPL)-Ed., 5/12/14 NOTE: Extended to 3/31/2014 per PI (Ed., 10/21/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhong, Fang (Jet Propulsion Laboratory) Beysens, Daniel (Commissariat a l'Energie Atomique at Grenoble and Ecole Supérieure de Physique et Chimie Industrielle) Garrabos, Yves (Centre National de la Recherche Scientifique (CNRS)-)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory) Lecoutre, Carole (Centre National de la Recherche Scientifique (CNRS)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15890		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologist, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning, Renormalization Group theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on “seemingly innocuous” assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient. NASA’s MISTE/COEX (microgravity scaling theory experiment/coexistence curve experiment) flight experiments were high precision studies of static critical phenomena planned on board the ISS. Those experiments were to address one of the most difficult experimental challenges in studying static properties of near critical fluid in microgravity, which is associated with the equilibration time of the sample. Recently, the DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques) facility supported by CNES/NASA was successfully integrated into the International Space Station (ISS). The DECLIC facility developed by CNES investigators allows precision studies of non-equilibrium phenomena near the liquid-gas critical point. Multiple science inserts for the DECLIC facility have been developed to study room temperature and high temperature critical fluids. These inserts are currently on-board ISS to study the boiling phenomena, phase separation and piston effect of near critical fluids. The DECLIC/ALI experiment is to be performed in the near future. Detailed discussions with the DECLIC PI’s have clearly indicated that additional important scientific investigations can be carried out utilizing the available DECLIC/ALI hardware. Taking advantage of the current CNES funded DECLIC facility, we propose to NASA to study the temperature and density relaxation behavior of the fluid (SF6) near its liquid-gas critical point by utilizing DECLIC/ALI experiments. By analyzing experimental data taken under microgravity conditions, critical data to understand the equilibration processes of a near critical fluid, especially in the two-phase region, can be obtained that is not feasible on the Earth. The results of data analysis, modeling, simulations, and ground support experiments will enhance the scientific return of the proposed DECLIC investigations. This study of equilibration near a critical fluid will directly support the objectives of NASA’s MISTE/COEX flight experiment and benefit future critical phenomena studies utilizing the DECLIC facility. The research team of this proposal consists of NASA MISTE/COEX investigators and CNES DECLIC principal investigators.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide an unique opportunity to measure experimental properties of a fluid near its liquid-gas critical point without gravitational effects and to refine the theoretical equation of state models. The critical and supercritical state of fluids exhibit unique properties. For instance, the fluid near the critical point and supercritical state shows very large compressibility. The use of these fluids, for instance, for the storage of cryogenic propellant, under normal gravity and/or reduced gravity conditions raises fundamental questions regarding fluid dynamics, heat transfer, interfacial phenomenon. These experimental works will enhance the understanding of basic physics regarding the fluids near critical state. Potential impacts are fields of applications like supercritical fluid extraction, Dry-cleaning, nano and micro particle formation, supercritical drying, supercritical water oxidation.
Task Progress:	During the period of this report, DECLIC/ALI is operated to run a script to measure the turbidity of the sample. Although the flight experiments are currently on-going and further planned, the preliminary turbidity data has been analyzed and shown in various workshops. Theoretical progress has been made in comparison between theoretical crossover equation-of-state model (CPM) and the massive renormalization approach. Preliminary data analysis using the CPM model suggest that the average density of the cell is at 1.7% off-critical density. We are currently measuring more turbidity data to confirm the repeatibility of the data set. Theoretical effort is also being made to include the Green-Fisher exponent in the calculations of turbidity. During this period of report, the principal investigator also organized an annual DECLIC workshop at Jet propulsion Laboratory, Pasadena. Recent results from the ALI and other scientific investigations (HTI and DSI) of DECLIC facility were discussed, as well as future utilization plan for the DECLIC facility.
Bibliography Type:	Description: (Last Updated: 01/02/2025)
Abstracts for Journals and Proceedings	Lecoutre C, Garrabos Y, Beysens D, Hahn I. "Test of Ising-like equation-of-state models using turbidity measurements in near critical fluids. " 18th Symposium on Thermophysical Properties, Boulder, CO, June 24-29, 2012. 18th Symposium on Thermophysical Properties, Boulder, CO, June 24-29, 2012. PaperID 1132. http://thermosymposium.nist.gov/pdf/Abstract_1132.pdf , Jun-2012